Mathematical Writing

Personal Notes

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Chapter 1 General Principles

- Good writing is clear thinking made visible.
  Ambrose Bierce, Write it Right: A Little Blacklist of Literary Faults (1937)
- A writer needs three qualities: creativity, originality, clarity and a good short term memory.
  Desmond J. Higham, More Commandments of Good Writing (1992)

1.1 Writing helps you to learn

Writing is not simply a task to be done once research or other preparation is completed – it can be an integral part of the work process.

In writing, you clarify your own thoughts, and strengthen your conviction. Indeed, you probably grasp your thoughts for the first time.

Writing is a way of thinking. Writing actually creates thought, and generates your ability to think: you discover thoughts you hardly knew you had, and come to know what you know. You learn as you write.

1.2 Writing brings out gaps in your understanding

Writing brings out gaps in your understanding, by forcing you to focus on steps in your thinking that you might otherwise skip.

In this respect, writing is similar to explaining your ideas to a colleague.

Writing down partial research results as your work progresses can reveal structure and suggest further areas of development.

1.3 Good writing reflects clear thinking

It is very hard for a woolly thinker to produce clear prose.

Clear thinking leads to good organization, a vital ingredient of technical writing.

A well-organized piece is much easier to write than a badly organized one.

If you find a particular piece difficult to write it may be because you have not found the right structure to express your ideas.

1.4 Writing is difficult

Writing is one of the hardest things that people do.

It is often difficult to get started.

One solution, which works for some, is to force yourself to write something, however clumsy it may be, for it is often easier to modify something you have written previously than to compose from scratch.

1.5 To keep your prose simple and direct

The most fundamental tenet of technical writing is to keep your prose simple and direct.

Much of written English is unnecessarily complicated. In writing up your research you are aiming at a relatively small audience, so it is important not to alienate part of it with long-winded or imprecise text.

English may not be the first language of many of your readers – they, particularly, will appreciate plain writing. Aim for economy of words. Early drafts can usually be reduced in length substantially with consequent improvements in readability.

1.6 To receive, and learn from, constructive criticism

Probably the best way to improve your writing skills is to receive, and learn from, constructive criticism.

Ask a colleague to read and comment on your writing. Another reader can often find errors and suggest improvements that you miss because of your familiarity with the work. Criticism can be difficult to take, but it should be welcomed; it is a privilege to have someone else take the time to comment on your writing.

1.7 To read as much as you can

Another way to improve your writing is to read as much as you can, always with a critical eye.

In particular, the following mathematical books are recommended, written by outstanding mathematical writers:

- Forman S. Acton (1970), *Numerical Methods That Work*
- Albert H. Beiler (1966), *Recreations in the Theory of Numbers*
- David M. Burton (1980), *Elementary Number Theory*
- Gene H. Golub and Charles F. Van Loan (1996), *Matrix Computations*
- Paul R. Halmos (1982), *A Hilbert Space Problem Book*
- Donald E. Knuth (1973–1981), *The Art of Computer Programming*
- Beresford N. Parlett (1998), *The Symmetric Eigenvalue Problem*
- G. W. Stewart (1973), *Introduction to Matrix Computations*
- Gilbert Strang (1986), *Introduction to Applied Mathematics*

1.8 Expository writing in mathematics that have won prizes

Also worth studying are papers or books that have won prizes for expository writing in mathematics. These include prizes such as the Chauvenet Prize, the Lester R. Ford Award, the George Polya Award, the Carl B. Allendoerfer Award, the Beckenbach Book Prize, and the Merten M. Hasse Prize.

Chapter 2 Mathematical Writing

- Suppose you want to teach the "cat" concept to a very young child. Do you explain that a cat is a relatively small, primarily carnivorous mammal with retractile claws, a distinctive sonic output, etc.? I'll bet not. You probably show the kid a lot of different cats, saying "kitty" each time, until it gets the idea. To put it more generally, generalizations are best made by abstraction from experience.
  — Ralph P. Boas
- A good notation should be unambiguous, pregnant, easy to remember; it should avoid harmful second meanings, and take advantage of useful second meanings; the order and connection of signs should suggest the order and connection of things.
  — George Pólya

2.1 What is a Theorem?

The differences between theorems, lemmas, and propositions depend to some extent on the context in which a result appears.

Generally, a **theorem** is a major result of independent interest. Its proof is usually nontrivial.

A **lemma** is an auxiliary result — a stepping stone toward a theorem. Its proof may be easy or difficult. A straightforward and independent result worth encapsulating but not meriting the title of theorem may also be called a lemma. Some lemmas are very famous, such as the Riemann-Lebesgue Lemma and Farkas's Lemma.

Whether a result should be formally stated as a lemma or mentioned informally depends on the level of the writing. It is not advisable to label all results as theorems, because this obscures the logical structure and fails to emphasize the most important outcomes. When in doubt, use **lemma**.

The term **proposition** is less common and less clearly defined. It typically denotes a minor or auxiliary theorem. Note that a proposition is *not* "a theorem that might not be true".

A **corollary** is a direct or easy consequence of a lemma, theorem, or proposition. One must distinguish clearly between a corollary and a generalization of a result. Do not over-glorify a corollary by omitting its label, as this distorts the logical hierarchy.

A **conjecture** is a statement the author believes to be true but cannot prove or disprove. A **hypothesis** is a statement assumed as a basis for further reasoning, such as an induction hypothesis.

2.2 Proofs

Readers are often not interested in full details of a proof, but want to understand the outline and key ideas. They hope to learn a technique or principle applicable elsewhere.

To help readers, emphasize the structure of a proof, the difficulty of each step, and the key ideas that make it work. Use guiding phrases such as:

- The aim/idea is to …
- Our first goal is to show that …
- Now for the harder part.
- The trick of the proof is to find …
- The essential observation is that …

- It remains to show that …

When omitting part of a proof, indicate the nature and length of the omission:

- It is easy to show that …
- Some tedious manipulation yields …
- An easy induction gives …
- An argument similar to that in …shows that …

The end of a proof is usually marked by the Halmos symbol □, or occasionally by QED.

2.3 The Role of Examples

A highly effective pedagogical tactic is to discuss specific examples *before* the general case. Though many mathematicians prefer to start general, beginning with examples is often clearer.

Good examples engage readers and build intuition. Exercises in textbooks are also examples; they must be carefully chosen to be purposeful and illustrative. The first few examples and exercises should be strong enough to gain the reader's confidence.

Avoid hiding key information in exercises or making them unnecessarily difficult. The level and purpose of examples must match the intended audience.

2.4 Definitions

When formulating a definition, ask three questions: **why?**, **where?**, and **how?**

First, ask whether the definition is really necessary. Too many unnecessary definitions complicate the writing and burden the reader. Use descriptive phrases instead of new terms when appropriate.

Second, decide where to place the definition. Avoid long lists of definitions at the beginning. Ideally, a definition should appear just before the term is first used. If the term is reused after some pages, reinforce it with redundancy, e.g., "the optimal step length α^*".

Third, decide how to define the term. Aim for a definition that is:

- short and clear,
- based on a fundamental property,
- consistent with related definitions.

For standard notation, clarify any potential ambiguity, e.g., "the spectral radius $\rho(A)$".

2.5 Notation

Good notation is essential for clear mathematical writing. It should be:

- unambiguous,
- easy to remember,
- consistent throughout the work,
- suggestive of the underlying meaning.

Avoid overloading symbols or using conflicting notation. Do not introduce unnecessary notation.

2.6 Words versus Symbols

Balance the use of words and mathematical symbols. Do not over-rely on symbols at the expense of readable prose. Sentences should be grammatical whether or not formulas are included. Use consistent grammatical structure when presenting parallel ideas.

2.7 Displaying Equations

Display important equations on separate lines. Make displayed formulas part of the sentence structure, with appropriate punctuation. Keep displayed expressions as simple as possible, and align related equations clearly.

2.8 Parallelism

Use parallel structure for related ideas, similar conditions, or analogous cases. Parallelism improves readability and emphasizes logical relationships.

2.9 Dos and Don'ts

- Do label results appropriately (theorem, lemma, proposition, corollary).
- Do emphasize key ideas and structure in proofs.
- Do use examples to motivate and clarify general theory.
- Do place definitions near their first use.
- Do keep notation clear, consistent, and minimal.
- Don't overuse theorems or mislabel corollaries.
- Don't omit guiding explanations in proofs.
- Don't overload readers with unnecessary definitions or symbols.

Chapter 3 Writing a Paper

3.1 Title

According to Kerut, for every person who reads the whole text of a scientific paper, five hundred read only the title. This statistic emphasizes the importance of the title. The title should give a terse description of the content, to help someone carefully scanning a journal contents page or a reference list decide whether to read the abstract or the paper itself. Ideally, it should also be catchy enough to attract the attention of a browser. Achieving a balance between these two aims is the key to writing an effective title.

The title "A note on the interpolation problem" is too vague: what is the breakthrough heralded by "A note", and which interpolation problem is under discussion? Similarly, "Approximation by cubic splines" is too vague, since the approximation problem being addressed is not clear.

If the title is to be capitalized then all words except articles, short prepositions and conjunctions should be capitalized.

Rules of thumb are that a phrase should not be split between lines, a line should not start with a weak word such as a conjunction, and the lines should not differ too much in length.

3.2 Author List

In 1940, over 90% of papers reviewed in Mathematical Reviews had one author. Today that figure is about 50%, showing that the proportion of jointly authored works in mathematics has increased greatly. There are no hard and fast rules about the order in which the authors of a multiply authored paper are listed. Sometimes the person who did the greatest part of the work is listed first. Sometimes the academically senior person is listed first. Perhaps most often, the authors are listed alphabetically, which is the practice I favour.

Make sure that you use precisely the same name on each of your publications.

3.3 Date

Always date your work. If you give an unpublished paper to others they will want to know when it was written. You may not be able to distinguish different drafts if they are not dated. The date is usually placed on a line by itself after the author's name, or in a footnote. Spell the month out, rather than using numerical forms, since European and American authors interpret dates in opposite senses.

3.4 Abstract

The purpose of the abstract is to summarize the contents of the paper. It should do so in enough detail to enable the reader to decide whether to read the whole paper. The reader should not have to refer to the paper to understand the abstract.

Many mathematics journals state a maximum size for the abstract, usually between 200 and 300 words. The shorter it is, the better, subject to the constraint of it being sufficiently informative. Most abstracts occupy one paragraph.

- Avoid mathematical equations in the abstract if possible, particularly displayed equations.

- Do not cite references in the abstract.
- Try to make the abstract easy to understand for those whose first language is not English.
- The abstract should not make claims that are not justified in the paper.
- The abstract should give some indication of the conclusions of the paper.
- Your abstract should lay claim to some new results, unless the paper is a survey.
- Try not to start the abstract with "In this paper" or "This paper".

3.5 Key Words and Subject Classifications

Some journals list key words supplied by the author, usually after the abstract. The number of key words is usually ten or less. Since the key words may be used in computer searches, you should try to anticipate words for which a reader might search and make them specific enough to give a good indication of the paper's content.

Some journals also require subject classifications, such as the AMS Mathematics Subject Classifications.

3.6 The Introduction

Perhaps the worst way to begin a paper is with a list of notation or definitions. It can be argued that the first sentence should be the best in the paper – its job is to entice the uncommitted reader into reading the whole paper.

An introduction is fairly short. It should define the problem, explain what the work attempts to do, and outline the plan of attack. Unless the paper is very short it is advisable to outline its organization towards the end of the introduction.

3.7 Computational Experiments

Many papers describe computational experiments. These may be done for several reasons: to gain insight into a method, to compare competing methods, to verify theoretical predictions, to tune parameters in algorithms and codes, and to measure the performance of software.

When you report the results of a computational experiment you should give enough detail to enable the results to be interpreted and the experiment to be reproduced. Error measures should be normalized.

One of the difficulties in designing experiments is finding good test problems – ones which reveal extremes of behaviour, cover a wide range of difficulty, are representative of practical problems, and have known solutions.

3.8 Tables

To maximize readability the table design should be as simple as possible. Repetition should be avoided; for example, units of measurement or descriptions common to each entry in a column should go in the column header. Only essential information should be included in a table. The caption should be informative and should not merely repeat information contained in the table.

3.9 Citations

The two main styles of citation in mathematics journals are by number and by name and year. It is good style to incorporate the author's name if the citation is more than just a passing one.

When you cite several references together it is best to arrange them so that the citation numbers are in increasing order. It is important to be aware that the reference list says a lot about a paper.

For one or two authors, both names are given. If there are three authors, all three are listed in the first citation and subsequent citations use "et al."

3.10 Conclusions

If there is a conclusions section it should not simply repeat earlier sections in the same words. It should offer another viewpoint, discuss limitations of the work, or give suggestions for further research.

3.11 Acknowledgements

Be sure to acknowledge any financial support for your work: grants, fellowships, studentships, sponsorship. It is customary to thank anyone who read the manuscript in draft form and offered significant suggestions for improvement.

3.12 Appendix

An appendix contains information that is essential to the paper but does not fit comfortably into the body of the text. The most common use of an appendix is to present detailed analysis that would distract the reader if included in the main text.

3.13 Reference List

Preparing the reference list can be one of the most tedious aspects of writing a paper. The precise format in which references are presented varies among publishers. Do not rely on secondary sources – always check the original reference. Make sure that every reference is actually cited in the paper.

3.14 Numbering Mathematical Objects

Generally, you should number only those equations that are referenced within the text. Except in very short papers it is best to number equations by section rather than globally. The same applies to the numbering of theorems and other mathematical objects.

3.15 Plagiarism

Plagiarism is the act of publishing borrowed ideas or words as though they are your own. It is a major academic sin. If you copy a sentence or more you should either place it in quotes and acknowledge the source, or give an explicit reference.

Self-plagiarism means reworking what you yourself have previously published without citing the source, thus passing it off as new.

Chapter 4 Revising a Draft

- Simply go through what you have written and try to curb the length of sentences, question every passive verb and if possible make it active, prune redundant words, and look for nouns used instead of verbs.
  Bernard Dixon, Sciwrite (1973)
- Every single word that I publish I write at least six times.
  Paul R. Halmos
- When one has finished a substantial paper there is commonly a mood in which it seems that there is really nothing in it. Do not worry, later on you will be thinking "At least I could do something good then."
  J. E. Littlewood
- Although they look like a first draft, they had already been rewritten and retyped—like almost every other page—four or five times. With each rewrite I try to make what I have written tighter, stronger and more precise, eliminating every element that is not doing useful work. Then I go over it once more, reading it aloud, and am always amazed at how much clutter can still be cut.
  William Zinsser, On Writing Well

4.1 How to Revise

All writing benefits from revision. Your first attempt can always be made clearer, more concise, and more forceful. Effective revision is a skill that is acquired through practice.

Put a draft aside for a few days, so that you can examine it with a fresh mind. Then analyse the draft in different ways. Read it aloud and listen to the rhythm. Read it at high speed. Does the text flow? Are the sentences of varying length? Is there a good balance between equations and text? Does the paper look inviting? Do the key ideas stand out?

Many readers will skim over formulas on their first reading of your exposition. Therefore, your sentences should flow smoothly when all but the simplest formulas are replaced by "blah" or some other grunting noise.

Some writers adopt the spiral technique of writing, whereby sections or chapters are written and revised in the order 1, 2, 1, 2, 3, 1, 2, 3, 4, ... This is described as the best way to start writing, perhaps the only way.

But it is not always necessary to start at the beginning: for example, you can leave the introduction to last and begin by writing a numerical examples section or your conclusions.

I make no distinction between editing, revising and rewriting in my own work. I begin with the intention of making whatever changes are necessary to produce the best exposition I can. This may involve minor deletions, insertions and reorderings, or it may require complete rewrites of paragraphs and sections.

Although I made full use of computers in writing this book, I did most of my writing and revising at my desk by writing on a blank sheet or a printout of a draft; only occasionally did I compose at the keyboard. When marking corrections, I use a different colour for each pass through the document and my printouts usually finish up covered in coloured ink.

Write or print your drafts with wide margins and with a line spacing big enough for you to write between the lines. This will give you a lot of space to mark revisions, which in turn will encourage you to make them.

4.2 Checklist for Revising

1. Can any words, phrases or sentences be deleted without loss of force or meaning (every such unit must earn its place)?
2. Can you replace any long, woolly words with short, pithy ones (for example, require → need, utilize → use)?
3. Are sentences and paragraphs in the best order, or should they be reordered or restructured?
4. Are there any ambiguities or blatantly wrong words (such as factor instead of feature)?
5. Is there any unnecessary repetition?
6. Can you convert a sentence from the passive to the active voice?
7. Is every claim fully supported?
8. Are the mathematical arguments and equations correct?
9. Is the notation consistent? Can it be simplified or made more logical?
10. Have quotations, references and numerical results been copied into the paper correctly?
11. Is due reference made to the work of other authors (beware of "citation amnesia")?
12. Are equations, results and the reference list properly numbered?
13. Are all the cross-references and bibliographic citations correct?

4.3 Examples

Abstract—This paper discusses the aims and methods of the FTEsol project and in this context discusses the architecture and design of the control system being produced as the focus of the project.

The phrases in this context and as the focus of the project serve no useful purpose and can be deleted. It is not a good idea to begin an abstract with This paper. A complete rewrite is needed: "The aims and methods of the FTEsol project are discussed, together with the architecture and design of the control system being produced." This version avoids the repeated discusses in the original.

Theorem 4.1

Let $A \in \mathbb{R}^{m\times n}$ be a given matrix, and let $A = U\Sigma V^T$ be a singular value decomposition of A. Then the problem

$$\max\{\operatorname{Re}\operatorname{trace}(AQ) : Q \in \mathbb{R}^{n\times n} \text{ is orthogonal}\}$$

has the solution $Q = VU^T$, and the value of the maximum is $\sigma_1(A) + \cdots + \sigma_n(A)$, where $\sigma_i(A)$ are the singular values of A. ♡

Revised theorem 4.1.

Theorem 4.2

Let $A \in \mathbb{R}^{m\times n}$ have the singular value decomposition $A = U\Sigma V^T$. Then

$$\max\{\operatorname{Re}\operatorname{trace}(AQ) : Q \in \mathbb{R}^{n\times n} \text{ is orthogonal}\} = \sum_{i=1}^{n} \sigma_i,$$

where the σ_i are the singular values of A, and the maximum is attained when $Q = VU^T$. ♡

Revised theorem 4.2.

Theorem 4.3

Let $A \in \mathbb{R}^{m\times n}$ *have the singular value decomposition* $A = U\Sigma V^T$ *with singular values* σ_i $(i = 1, \ldots, n)$*, where* $\Sigma = \operatorname{diag}(\sigma_i)$*. Then*

$$\max\left\{\operatorname{Re}\operatorname{trace}(AQ) : Q \in \mathbb{R}^{n\times n} \text{ is orthogonal}\right\} = \sum_{i=1}^{n} \sigma_i$$

and the maximum is attained when $Q = VU^T$*.* ♡